

Limitation of the Support-Confidence Framework

- Are s and c interesting in association rules: " $A \Rightarrow B$ " [s, c]? **Be careful!**
- Example: Suppose one school may have the following statistics on # of students who may play basketball and/or eat cereal:

	play-basketball	not play-basketball	sum (row)
eat-cereal	400	350	750
not eat-cereal	200	50	250
sum(col.)	600	400	1000

2-way contingency table

- Association rule mining may generate the following:
 - $play\text{-}basketball \Rightarrow eat\text{-}cereal$ [40%, 66.7%] (higher s & c)
- But this strong association rule is misleading: The overall % of students eating cereal is 75% > 66.7%, a more telling rule:
 - $\neg play\text{-}basketball \Rightarrow eat\text{-}cereal$ [35%, 87.5%] (high s & c)